The Genus of Cayley Graphs on Billiard Surfaces

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Abstract

In this article we discuss a connection between two well-known constructions in mathematics: Cayley graphs and rational billiard surfaces. We describe a natural way to draw a Cayley graph of a dihedral group on each rational billiard surface. Both of these objects have the concept of "genus" attached to them. For the Cayley graph, the genus is defined to be the lowest genus amongst all surfaces that the graph can be drawn on without edge crossings. We prove that the genus of a Cayley graph associated with a triangular billiard table is always zero or one. One reason this is interesting is that there exist triangular billiard surfaces of arbitrarily high genus, so the genus of the associated graph is often much lower than the genus of the billiard surface.

1 Rational Billiard Surfaces

The *rational polygonal billiard surface* is a famous construction in topological dynamics. See [MT02] for an excellent survey. Although billiard surfaces have served both as motivation for and examples of recent advances in sophisticated mathematics such as the work of Mirzhakani on moduli spaces [Wri20], they have an intuitive construction. When following the path of a point mass bouncing around inside a polygon, when the point strikes a wall of the polygon we continue its path in a straight line in a reflected copy of the polygonal table. See Figure 1.

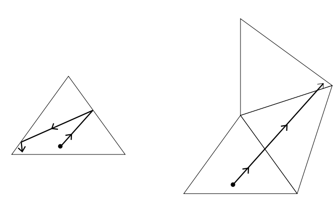

Figure 1: Unfolding a billiard path in a triangle with angles $\frac{3\pi}{10}, \frac{3\pi}{10}, \frac{4\pi}{10}$.

Copies of the table that correspond to identical directions of the billiard ball are identified. This is known as "unfolding" the path. In this way, a billiard path can be viewed as a straight line on a compact surface instead of as a collection of line segments of various slopes. The resulting line has finite length if and only if it "closes up" – this occurs when the billiard ball returns to its initial position while traveling in its initial direction.

Here is a more rigorous description of a rational polygonal billiards surface. Begin with a polygonal region P in the plane whose internal angles are rational multiples of π. Consider the set of reflections

$R_1, \dots R_m$ across the lines L_i through the m sides of P. Such a reflection is an affine transformation of $\mathbb{R}^2$, and hence can be expressed as the composition of a linear transformation and a translation, as follows. Let L_i' be the line through the origin parallel to L_i, and let r_i be the linear transformation that is reflection across L_i'. Finally, let $\vec{t_i}$ be the vector orthogonal to L_i such that $L_i' + \vec{t_i} = L_i$. Then we can write R_i as the affine transformation $R_i(\vec{x}) = r_i(\vec{x}) + 2\vec{t_i}$. The transformation r_i is called the *derivative* of the transformation R_i. Observe that $R_j R_i(\vec{x}) = R_j(r_i(\vec{x}) + 2\vec{t_i}) = r_j(r_i(\vec{x}) + 2\vec{t_i}) + 2t_j = r_j r_i(\vec{x}) + 2r_j(\vec{t_i}) + 2\vec{t_j}$. The derivative of this last expression is $r_j(r_i(\vec{x}))$, so we see that the derivative of the composition of two reflections is the composition of the derivatives.

As a concrete example, suppose we impose coordinates on a right isosceles triangle by placing its vertices at the points (0,0),(1,0), and (0,1). Let L be the line which coincides with the triangle's hypotenuse and let L' be the line parallel to L which contains the origin. Let $R(\vec{x})$ be reflection across the line L and let $r(\vec{x})$ be the derivative of $R(\vec{x})$. Then $r(\vec{x})$ is reflection across L', and $R(\vec{x}) = r(\vec{x}) + 2\vec{t}$, where $\vec{t} =< \frac{1}{2}, \frac{1}{2} >$. See Figure 2 for an illustration of the actions of R and r on some point $\vec{x_0}$ within the triangle.

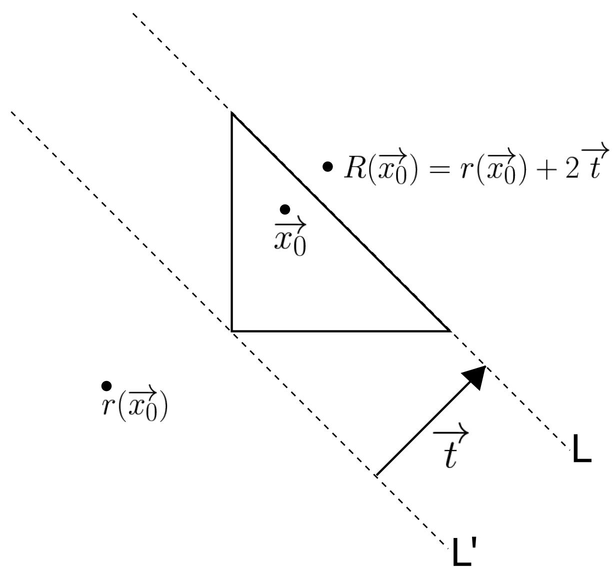

Figure 2: Reflection $R(x)$ across the hypotenuse of a $\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{4}$ triangle, along with its derivative $r(x)$.

These derivatives $r_1, \dots, r_m$ generate a group Γ under composition that consists of reflections across the origin (compositions of odd numbers of reflections) and rotations about the origin (compositions of even numbers of reflections). Suppose that the internal angles of P are $\alpha_1, \dots, \alpha_m$, where $\alpha_i = \dfrac{p_i \pi}{m}$ and $n, p_1, \dots, p_m \in \mathbb{Z}$. Further, suppose that n is the "least common denominator" in the sense that $\gcd(n, p_1, \dots, p_m) = 1$. In fact Γ is D_n, the dihedral group of order $2n$. We prove this fact in Lemma 1.

For this lemma, it is helpful to list some basic geometric formulas. We write $Rot(\theta)$ to denote Euclidean rotation about the origin by θ and $Ref(\theta)$ to denote reflection across the line through the origin that makes an angle of θ with the positive horizontal axis. Then the following formulas hold:

$$Ref(\theta_1)Ref(\theta_2) = Rot\left(2[\theta_1 - \theta_2]\right) \tag{1}$$

$$Ref(\theta_1)Rot(\theta_2) = Ref\left(\theta_1 - \frac{1}{2}\theta_2\right) \tag{2}$$

$$Rot(\theta_1)Ref(\theta_2) = Ref\left(\theta_2 + \frac{1}{2}\theta_1\right) \tag{3}$$

Lemma 1 *The group* Γ *generated by the derivatives of reflections in the sides of the polygon is* D_n*, the dihedral group of order* $2n$ *.*

Proof. Since n is the least common denominator of the reflections generating Γ, it follows from Equations 1-3 that any reflection in Γ has an angle that is an integer multiple of $\frac{\pi}{n}$, and any rotation in Γ has an angle that is an integer multiple of $\frac{2\pi}{n}$. Recall that D_n is the group which consists of precisely all of these reflections and rotations; therefore $\Gamma \subseteq D_n$.

Let r_i and r_{i+1} be reflections across the consecutive sides of our polygon which form the angle α_i. Equation 1 gives that $Rot\left(\frac{2p_i\pi}{n}\right)$ is in Γ for each i, since $r_{i+1}r_i = Rot(2\alpha_i) = Rot\left(\frac{2p_i\pi}{n}\right)$.

Since $\gcd(p_1, \ldots, p_m, n) = 1$, there exist integers $t_1, \ldots, t_m, s$ such that $\sum_{i=1}^m t_ip_i = 1 + ns$.

Thus

$$\sum_{i=1}^{m} t_i \frac{2p_i\pi}{n} = \frac{2\pi}{n} + 2\pi s$$

from which it follows that

$$\prod_{i=1}^{m} Rot\left(\frac{2p_i\pi}{n}\right)^{t_i} = Rot\left(\frac{2\pi}{n}\right) Rot(2\pi s) = Rot\left(\frac{2\pi}{n}\right)$$

and so $Rot\left(\frac{2\pi}{n}\right)$ is in Γ.

Therefore the set of rotations in Γ consists of the n rotations of the form $Rot(\frac{2k\pi}{n})$, $k = 0, \ldots, n-1$.

Finally, take some integer l such that there is a reflection in Γ with angle $\frac{l\pi}{n}$. Then by Equation 3, the reflection $Rot(\frac{2k\pi}{n})Ref\left(\frac{l\pi}{n}\right) = Ref\left(\frac{(l+k)\pi}{n}\right)$ is in Γ for each integer k. Therefore the set of reflections in Γ is the n reflections which have angles that are integer multiples of $\frac{\pi}{n}$. Hence, $\Gamma = D_n$.

■

Next, we consider the set $S = \{\sigma P : \sigma \in \Gamma\}$ of the $2n$ copies of P transformed by elements of Γ. We construct the rational billiards surface for P by "gluing" two copies together along a side if one copy can be seen as a mirror image of the other, reflected across that side. Specifically, for each $\sigma_j \in \Gamma$ and each r_i, we glue together $\sigma_j r_i R$ and $\sigma_j R$ along the "shared side". This is because, if e_i is the side of P corresponding to r_i then reflection across $\sigma(e_i)$ is equivalent to $\sigma r_i \sigma^{-1}$, and so the reflection across $\sigma(e_i)$ of $\sigma(R)$ is $\sigma r_i \sigma^{-1}(\sigma R) = \sigma r_i R$.

The result is a closed flat surface known as a *billiard surface*. A billiard surface is an example of a *translation surface* since, viewed as complex manifold with flat structure, its change-of-coordinate maps (away from a finite number of singular points located at the corners of the copies of P) are Euclidean translations. [MT02]

In this paper we will focus on billiard surfaces arising from rational triangles. Let $T(p_1, p_2, p_3)$ denote a rational triangle with internal angles $\alpha_i = \frac{p_i\pi}{n}$, where $n = p_1 + p_2 + p_3$ and $\gcd(p_1, p_2, p_3) = 1$. Let $X(p_1, p_2, p_3)$ denote the billiard surface arising from billiards in $T(p_1, p_2, p_3)$. See Figure 3 for a diagram of $X(3, 3, 4)$. The numerical labels indicate side identifications. This construction has been described as far back as 1936 by Fox and Kershner in [FK36].

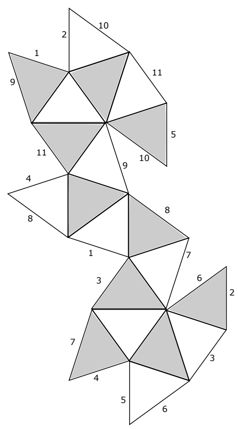

Figure 3: The billiard surface $X(3, 3, 4)$.

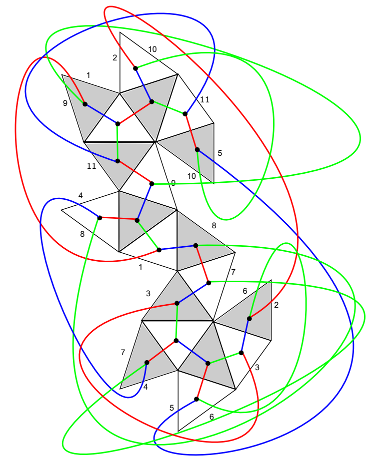

Figure 4: The graph $G(3, 3, 4)$ drawn on $X(3, 3, 4)$.

2 Cayley Graphs on Billiard Surfaces

2.1 Drawing the Graph

Next, we will make a connection between billiard surfaces and graphs. We draw a graph $G(p_1, p_2, p_3)$ on a triangular billiard surface $X(p_1, p_2, p_3)$ by drawing a vertex in the center of each triangle and connecting two vertices with an edge if and only if the vertices lie inside triangles which share a side. See Figure 4. Because each copy of $T(p_1, p_2, p_3)$ in $X(p_1, p_2, p_3)$ is associated with a unique element of D_n, we can identify the vertices of the graph with elements of D_n. We choose one initial copy of $T(p_1, p_2, p_3)$ and denote it eT, where e denotes the identity element of D_n. Let a, b, and c denote reflections across lines parallel to sides opposite α_1, α_2, and α_3, respectively. See Figure 6. Then the triangle eT is adjacent to triangles aT, bT, and cT, so we have edges connecting the vertex e to the vertices a, b, and c. Observe that if $\sigma \in D_n$ then the reflection of σT across the side of σT opposite the α_1 angle of σT is $\sigma a \sigma^{-1}(\sigma T) = \sigma a T$. This works the same way for the other two sides of the triangle. Hence, in $G(p_1, p_2, p_3)$, the vertex σ is connected by edges to σa, σb, and σc. See Figure 5.

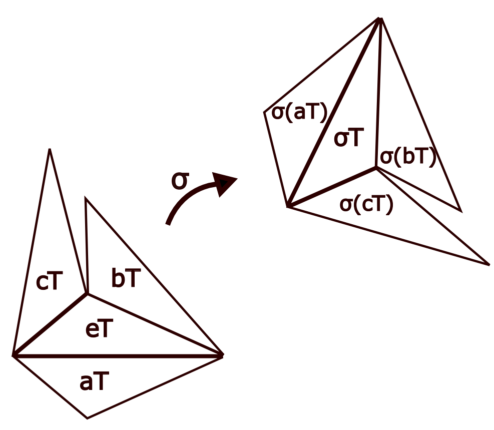

Figure 5: The action of some $\sigma \in D_n$ on part of the billiard surface.

In fact this graph is an example of a *Cayley graph*. We shall review the necessary background for Cayley graphs in the next section. Our explicit drawing of a Cayley graph on a billiard surface appears to be original. However, it should be noted that there is a similar concept – the *dessin d'* *enfant* – which has been widely studied for billiard surfaces. See [AI88]. The dessin can be viewed as the vertex-face dual of our graph, with extra structure added.

2.2 Notation and Basic Formulas

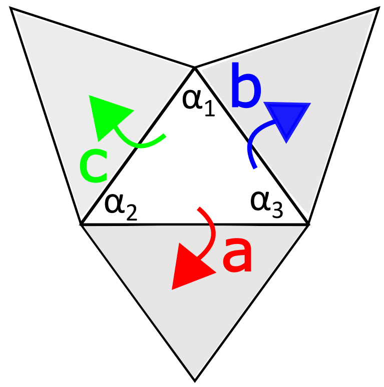

Figure 6: The reflections that generate $Cay(\{a, b, c\}, D_n)$.

Remark 1 *Suppose that our triangle is as in Figure 6, oriented so that the base of the triangle is parallel to the horizontal axis. It follows from the elementary properties of Euclidean reflections and Euclidean rotations listed above that:*

1. *ab is rotation by* $2\alpha_3$
2. *ac is rotation by* $-2\alpha_2$
3. *aba is reflection across a line making angle* α_3 *with the positive horizontal axis.*

A billiard path on $T(p_1, p_2, p_3)$ is called *closed* if the billiard ball returns to its initial position, traveling in its initial direction. There is a well-known correspondence between closed billiard paths on $T(p_1, p_2, p_3)$, cylinders on $X(p_1, p_2, p_3)$ and circuits in $G(p_1, p_2, p_3)$ [MT02]. Specifically, every infinite family of parallel equal-length closed billiard paths in $T(p_1, p_2, p_3)$ unfolds to a family of parallel finite geodesics on $X(p_1, p_2, p_3)$, forming a topological cylinder. The sequence of sides of T struck by each path in this family determines a sequence of reflections in $\Gamma(p_1, p_2, p_3)$, which form a circuit in $G(p_1, p_2, p_3)$.

For example, every right triangle has an infinite family of billiard paths consisting of striking four walls each; this corresponds to the closed circuit *abcb* in the corresponding Cayley graph (see Lemma 3). Similarly, the length six circuit identified in Theorem 6 corresponds to the family of doubles of the well-known "Fagnano orbit" that exists as a closed billiard path in any acute triangle.

See Figure 3 and Figure 4 for an example of a billiard surface and its accompanying Cayley graph. Although the genus of $X(3, 3, 4)$ is four, we shall show that the genus of $G(3, 3, 4)$ is zero.

3 Graph Theory Background

3.1 Cayley Graphs

Let Γ be a group. We say that a subset $S = \{g_1, g_2, ..., g_n\}$ of Γ is a *generating set* for Γ if every element of Γ can be expressed as a product of elements of S (and their inverses). We do not require S to be minimal – that is, we do not exclude the possibility that a proper subset of S may also be a generating set of Γ. The *Cayley graph* $Cay(S, \Gamma)$ of a group Γ with generating set $\{g_1, g_2, \ldots, g_n\}$ is a graph whose vertices are the elements of Γ, and whose edges represent multiplication by an element of the generating set. We draw an edge from x to y if $y = xg_i$ for some generator g_i. For example, in Figure 7 we see a drawing of the Cayley graph $Cay(\{a, b, c\}, D_3)$, where D_n is the dihedral group with $2n$ elements, and $\{a, b, c\}$ is the set of the three reflection elements in D_3. In graph theory, this graph is known as the *complete bipartite graph* $K_{3,3}$. A graph is *bipartite* if its vertex set can be partitioned into two subsets V_1 and V_2 such that each element of V_1 is adjacent only to elements of V_2. All the Cayley graphs in this paper are bipartite with Note that in general a Cayley graph is a directed graph. However, because the generating elements we use in this paper all have order 2, we replace the pairs of oppositely directed edges with single undirected edges to form an undirected graph.

Recall that Lemma 1 shows that the reflections in the sides of a rational triangle generate D_n

A graph is *regular* if each of its vertices have the same degree (number of neighbors). If that common degree is k, we say that the graph is *k-regular.* The Cayley graphs we draw on triangular billiards surfaces are necessarily all 3-regular.

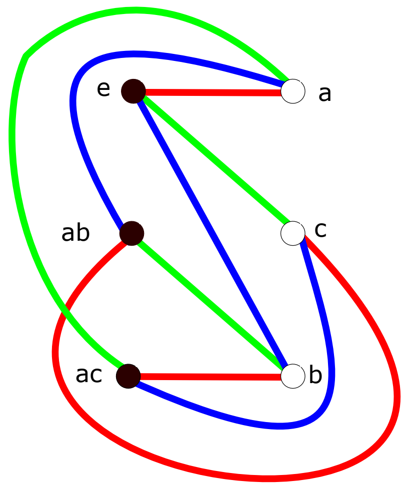

Figure 7: The Cayley graph for the (1,1,1) triangle is isomorphic to $K_{3,3}$.

3.2 Graph Genus and Graph Rotations

Definition 1 *The* genus *of a graph is the smallest non-negative integer g such that the graph can be drawn on a surface of genus g without edge crossings.*

A graph of genus zero is called *planar*. A classic theorem which can aid in computing graph genus is Kuratowksi's Theorem, which uses the concept of a *subdivision* of a graph. A subdivision of a graph G is a graph obtained from G by adding vertices of order 2 to some of the edges of G. See Figure 12 for an example of a graph which is a subdivision of $K_{3,3}$.

Theorem 2 *(Kuratowski) A graph G is not planar if and only if G contains a subgraph that is a subdivision of either the complete graph K_5 or of the complete bipartite graph $K_{3,3}$.*

Another tool for computing the genus of a graph is the concept of a *graph rotation.* The following definitions are from [HR03]. A *rotation of a vertex* is an ordered cyclic listing of the vertices adjacent to that vertex. A *graph rotation* consists of rotations of each vertex of the graph. This term is not to be confused with the concept of a Euclidean rotation of the plane.

A *circuit* of a graph is a sequence of vertices v_i and edges E_i of the form $v_0 E_1 v_1 E_2 v_2 \ldots E_n v_n$ such that $v_0 = v_n$ and such that for each i, E_i connects v_{i-1} to v_i. A circuit cannot use the same edge twice. In this article, since each undirected edge actually represents two directed edges, it is possible for a circuit to use the same edge twice, but not to use it twice in the same direction. We say that such a circuit has *length n*. A circuit may be represented by listing only the subsequence consisting of the vertices, and omitting the final vertex since it is equal to the first: $v_0 \dots v_{n-1}$. Or, it may be represented by listing only the edge subsequence: $E_1, \dots, E_n$. Finally, in a Cayley graph, where each edge is labeled with a generator of a group G, it can be of interest to list only the sequence of edge labels. This sequence can then be interpreted as a product of generators, which must equal the identity element in G.

Remark 3 *Observe that for any graph $G = Cay(\{a, b, c\}, D_n)$ where a, b, and c represent reflections:*

1. *$G = Cay(\{a, b, c\}, D_n)$ is defined since, by Lemma 1, $\{a, b, c\}$ does generate D_n.*
2. *Every circuit $v_0E_1v_1E_2v_2 \dots E_nv_n$ corresponds to a relation $E_1E_2 \dots E_n = R_0$, where R_0 is the identity element of D_n.*
3. *It follows from the previous observation that since each E_i is a Euclidean reflection and R_0 is a Euclidean rotation, circuits in G are always of even length. From a graph theoretic perspective, this is true because G is bipartite: the reflections and Euclidean rotations form the two bipartite sets.*

A graph rotation of a graph G induces a set of circuits on G such that each edge is traveled once in each direction. The circuits are obtained in the following way: the circuit which contains $...v_iv_j$ continues as $...v_iv_jv_k$, where v_k is the vertex directly following v_i in the rotation of v_j. This process of constructing a circuit terminates when we would be adding a directed edge to the circuit which already exists in that circuit. For example, consider the following rotation of $K_{3,3}$:

$$v_0.v_1v_3v_5$$
$$v_1.v_0v_4v_2$$
$$v_2.v_1v_5v_3$$
$$v_3.v_0v_2v_4$$
$$v_4.v_1v_3v_5$$
$$v_5.v_0v_2v_4$$

The notation $w.v_iv_jv_k$ means that the vertex w is adjacent to exactly the vertices v_i, v_j, v_k, and that the cyclic listing of these vertices is $v_iv_jv_k$.

The rotation in our example induces three circuits. This includes two circuits of length four: $v_0v_3v_2v_1$ and $v_1v_2v_5v_4$; and one circuit of length eight: $v_0v_1v_4v_3v_0v_5v_2v_3v_4v_5$. See Figure 8.

Let $r(\rho)$ denote the number of circuits induced by a rotation ρ of a graph G. So in the previous example, $r(\rho) = 3$. We call ρ a *maximal rotation* of G if $r(\rho) = \max_{\rho_i}\{r(\rho_i)\}$ where the ρ_i vary over all possible rotations of G.

The following formula provides the connection between graph genus and graph rotations.

Theorem 4 *[HR03] Let G be a connected graph with p vertices and q edges, and let ρ be a maximal rotation of G. Then the genus of G is g, where $p - q + r(\rho) = 2 - 2g$.*

Theorem 4 is related to Euler's characteristic formula, with circuits playing the role of the polygonal "faces" of the surface. With that analogy in mind, it is not surprising that a circuit induced by a graph rotation will never trace both directed edges of an edge consecutively. Indeed, this follows from the definition of a rotation of a vertex, since if a circuit includes an edge starting at v_i and ending at v_j,

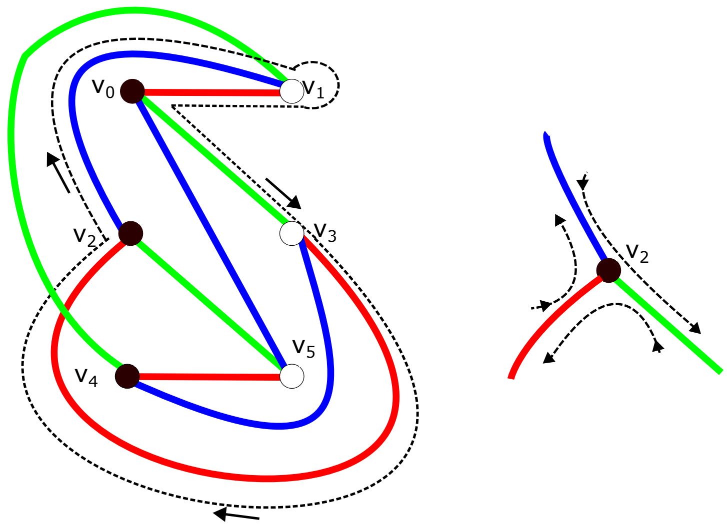

Figure 8: At left, the circuit $v_0v_3v_2v_1$ on $K_{3,3}$ induced by the graph rotation in our example. At right, the rotation of the vertex v_2 given in our example.

then the next edge of the circuit must start at v_j and end at some vertex $v_k \neq v_i$ in the cyclic ordering of the vertices adjacent to v_j.

As a second example, consider $G(1,3,3)$. See Figure 9. Here several vertices and edges are drawn twice; the reader should view identical objects as being identified. We color the "a", "b", and "c" edges red, green, and blue, respectively. To simplify graph labeling, we write $x = ba$. Here, we have that $ca = ab$ and also that $cb = abab$. Note that, since cba is a Euclidean reflection, $cbacba = (cba)^2 = e$. So following edges labeled a,b,c,a,b,c in that order will yield a length 6 circuit, no matter where we start in the graph. These circuits are indicated in Figure 9 by the circular arrows within each hexagon. Observe that each edge in $G(1,3,3)$ is traversed exactly twice by these circuits – once in each direction.

Next, consider Figure 10, wherein we have taken the graph in Figure 9 and redrawn it slightly so that the outer boundary forms a larger hexagon whose opposite sides are identified. Since it is well-known that identifying opposite sides of a hexagon creates a genus 1 surface, we see that $G(1,3,3)$ has genus at most 1.

4 Results for Triangles

The goal of this section is Theorem 6, which states that the genus of $G(p_1,p_2,p_3)$ is always zero or one. This result is in contrast to the fact that the surface $X(p_1,p_2,p_3)$ can have arbitrarily high genus [AI88].

We begin by generalizing the last example in the previous section.

Lemma 2 *The Cayley graph $G = G(p_1,p_2,p_3)$ has genus at most 1.*

Proof. Let $n = p_1 + p_2 + p_3$. Since D_n has $2n$ elements, G has $2n$ vertices. Since G is 3-regular, it follows that G has $\frac{3}{2}(2n) = 3n$ undirected edges. Let ρ be a maximal rotation of G. By Theorem 4, $2n - 3n + r(\rho) = 2 - 2g$, so $g = 1 + \dfrac{n - r(\rho)}{2}$.

Now consider the graph rotation ρ_1 defined on G in the following way. Each vertex x of G is adjacent to three other vertices $v_{x,a}, v_{x,b}, v_{x,c}$ via edges with labels a, b, and c, respectively. Using this notation, for each x, define the rotation ρ_1 at x to be $x.v_{x,a}v_{x,b}v_{x,c}$. That is, any circuit induced by ρ_1 which arrives at a vertex along an edge labeled a must next exit that vertex along an edge labeled

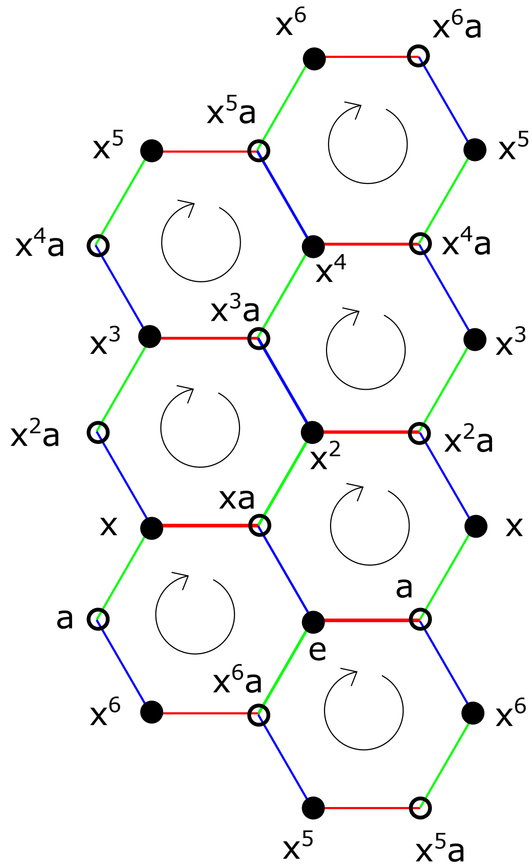

Figure 9: G(1,3,3), with "a" "b" and "c" edges colored red, blue and green respectively. Some edges and vertices are drawn twice. Circuits are indicated with circular arrows. We define $x = ba$.

b; if it arrives at a vertex along a b edge then it must leave along a c edge, and if it arrives along a c edge then it must leave along an a edge. It follows that any circuit induced by ρ_1 whose first edge is labeled a must begin with the edge string $abcabc$. Since the element $abc \in D_n$ is a Euclidean reflection, we know that it is its own inverse. Hence $abcabc = (abc)(abc)$ is the identity in D_n, from which it follows that $abcabc$ always describes a complete circuit in the directed graph. Indeed, any directed edge in G is a part of such a circuit. Since G has $6n$ directed edges, it follows that ρ_1 induces exactly $r(\rho_1) = \dfrac{6n}{6} = n$ circuits. Thus $g \leq 1 + \dfrac{n - r(\rho_1)}{2} = 1$. That is, the genus of G is at most 1. ■

Lemma 3 *$G = G(p_1, p_2, p_3)$ has a length four circuit if and only if $T = T(p_1, p_2, p_3)$ is either isosceles or a right triangle.*

Proof. Suppose that $G(p_1, p_2, p_3)$ has a length four circuit: call it $x_1x_2x_3x_4$, where each x_i is in the generating set $\{a, b, c\}$. The first two elements x_1, x_2 of the circuit combine to form a Euclidean rotation, as do the last two elements x_3, x_4. Furthermore, since any two sides of a triangle are adjacent, $x_1x_2 = Rot(\pm 2\alpha_i)$ and $x_3x_4 = Rot(\pm 2\alpha_j)$ for some internal angles of the triangle α_i and α_j. Since $x_1x_2x_3x_4$ is a circuit, it follows that $Rot(\pm 2\alpha_i)Rot(\pm 2\alpha_j) = Rot(0)$, and so $\pm 2\alpha_i \pm 2\alpha_j = 2k\pi$ for some integer k and some choice of sign on each angle. If $i = j$, it follows that $\alpha_i = \pi/2$, and hence T is a right triangle. If $i \neq j$, it follows that $\alpha_i = \alpha_j$ for distinct internal angles of T, implying that T is isosceles.

Conversely, suppose that $T(p_1, p_2, p_3)$ is isosceles. Without loss of generality say that $p_2 = p_3$. Then we see that $abac$ is the identity in Γ, so $abac$ is a length 4 circuit in G. Finally, suppose instead

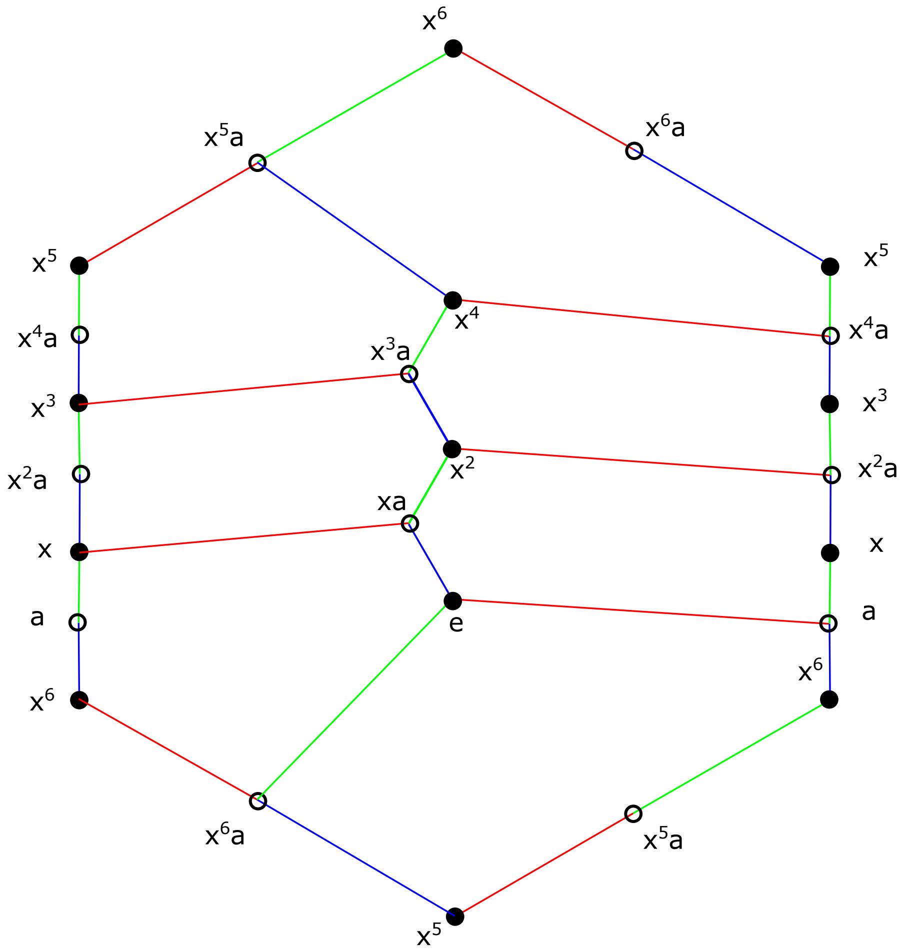

Figure 10: G(1,3,3), drawn on a hexagonal torus.

that $T(p_1, p_2, p_3)$ is a right triangle. Say $\alpha_3 = \dfrac{\pi}{2}$. Then since $ab = Rot(2\alpha_3) = Rot(\pi)$, it follows that the element $abab = Rot(\pi)Rot(\pi) = Rot(0)$ is the identity in Γ and hence the word $abab$ is a circuit in G. ■

Lemma 4 *If T is an isosceles or right rational triangle, then two of the three reflections generate D_n. In particular, if we label T so that α_2 is the right angle (if T is right) or so that α_2 and α_3 are the congruent angles (if T is isosceles), then D_n is generated by the set $\{a, b\}$.*

Proof. Suppose that $T = T(p_1, p_2, p_3)$, with $n = p_1 + p_2 + p_3$. We have that $ab = Rot\left(\dfrac{2p_3\pi}{n}\right)$. The order of ab is the smallest positive integer k such that:

$$(ab)^k = Rot(0)$$

$$Rot\left(\frac{2p_3k\pi}{n}\right) = Rot(0)$$

$$p_3k \equiv 0 \pmod{n}.$$

This implies that if $\gcd(p_3, n) = 1$ then the order of ab is n. In turn, if ab has order n then it generates an index 2 subgroup of D_n, and it follows that a and b together generate all of D_n.

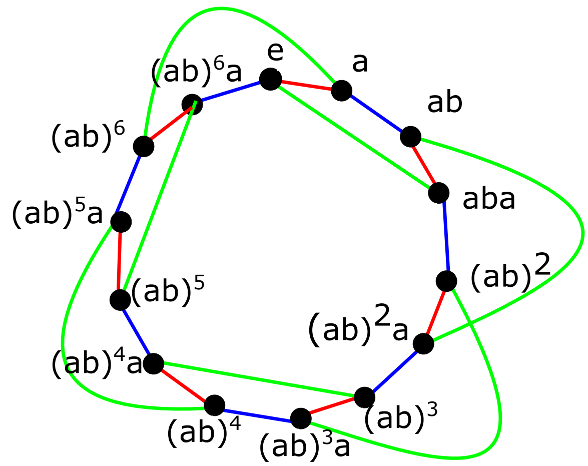

Figure 11: A drawing of $Cay(\{a, b, c\}, D_7)$.

Hence to prove the lemma it suffices to show that one of the p_i is relatively prime to n. First suppose that T is isosceles. Without loss of generality say that $p_2 = p_3$. Let $m = \gcd(n, p_3)$. Then since $p_1 = n - 2p_3$ we see that m divides p_1. But $\gcd(p_1, p_2, p_3) = 1$, so $m = 1$. Therefore p_3 and n are relatively prime.

Now suppose instead that T is a right triangle. Without loss of generality, say that α_2 is the right angle, so that $p_2 = \dfrac{n}{2}$ and $p_1 + p_3 = p_2$. Note that p_3 and p_1 cannot both be even, because if they were then p_2 would also be even, but $\gcd(p_1, p_2, p_3) = 1$. So let p_3 be odd and write $k = \gcd(n, p_3)$. Since p_3 is odd, k is odd. Note that $2p_1 = n - 2p_3$, so since k divides n and p_3, it must also divide $2p_1$. But k is odd, so in fact k divides p_1. Then k also divides $p_2 = p_1 + p_3$, and hence $k = 1$ since $\gcd(p_1, p_2, p_3) = 1$. Thus p_3 is relatively prime to n. ■

A *Hamiltonian cycle* in a graph is a path which visits each vertex of the graph exactly once, and such that the first and last vertices of the path are the same.

Corollary 5 *If T is a rational triangle that is isosceles or right, then its sides can be labeled so that $Cay(a, b, D_n)$ is a Hamiltonian cycle of $Cay(a, b, c, D_n)$.*

Proof. By Lemma 4, if T is isosceles or right then we can label its edges so that $\{a, b\}$ is a generating set for D_n, where $n = p_1 + p_2 + p_3$. Using this labeling, then $Cay(a, b, D_n)$ is a well-defined connected subgraph of $Cay(a, b, c, D_n)$ that contains all the vertices of $Cay(a, b, c, D_n)$. Furthermore, since each vertex of $Cay(a, b, D_n)$ has degree 2, $Cay(a, b, D_n)$ must be a Hamiltonian cycle of $Cay(a, b, D_n)$. ■

Note that it is not the case for *all* rational triangles that at least one of the p_i is relatively prime to n. For example, consider $T(5, 9, 16)$.

Lemma 5 *Let $G = G(p_1, p_2, p_3)$ be the Cayley graph for an isosceles triangle with interior angles $\dfrac{p_1\pi}{n}, \dfrac{p_2\pi}{n}, \dfrac{p_3\pi}{n}$. Then the genus of G is 0 if and only if n is even.*

Proof. Suppose that n is odd. We will show that the corresponding graph is not planar. Without loss of generality, we label our isosceles triangle as in Figure 6 so that $\alpha_2 = \alpha_3$. Then $c = aba$ and we have a Cayley graph such as the one depicted in Figure 11. Since α_3 is one of the two congruent angles of the triangle, Lemma 4 and Corollary 5 guarantee that the subgraph $Cay(\{a, b\}, D_n)$ consists of a Hamiltonian cycle of length $2n$. We choose two sets of three vertices: $V_1 = \{e, ab, (ab)^2\}$ and $V_2 = \{a, aba, (ab)^2a\}$ to be our bipartite sets. In the full graph G we already have edges connecting ab to each element of V_2; we also have edges connecting e to a and aba; and we have edges connecting $(ab)^2$ to aba

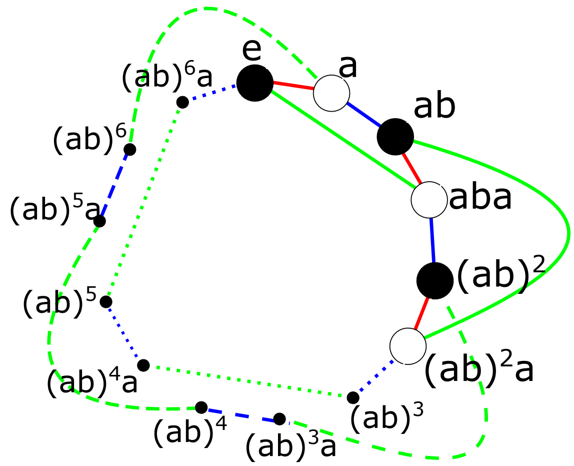

Figure 12: A subgraph of the Cayley graph for an isosceles triangle when $n = 7$, pictured as a subdivision of $K_{3,3}$. The dashed edges form a subdivision of a single edge of $K_{3,3}$, as do the dotted edges.

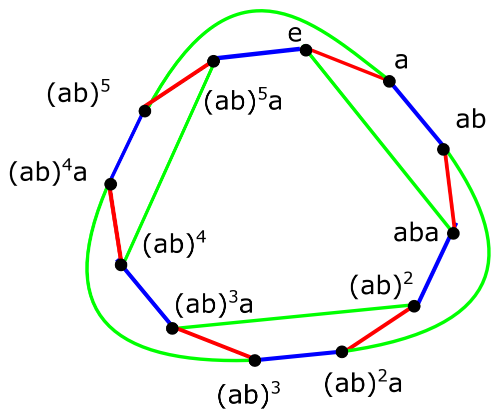

Figure 13: The Cayley graph for an isosceles triangle when $n = 6$.

and $(ab)^2a$. This leaves two edges to complete a subdivision of $K_{3,3}$. As exemplified in Figure 12, we can connect a to $(ab)^2$ via the sequence $a, (ab)^{n-1}, (ab)^{n-2}b, (ab)^{n-3}, (ab)^{n-4}a, \ldots, (ab)^3a, (ab)^2$. Similarly, we can connect e to $(ab)^2a$ via the sequence $e, (ab)^{n-1}a, (ab)^{n-2}a, (ab)^{n-3}a, (ab)^{n-4}, \ldots, (ab)^3, (ab)^2a$. Hence by Kuratowski's Theorem, the graph is not planar; that is, its genus is not zero.

Now suppose that n is even. We will show that the graph is planar. Since a and b generate D_n, we can draw a loop in the plane with all vertices of D_n on it, using all the a and b edges. It remains to draw all the c edges without creating any edge crossing. As exemplified in Figure 13, this can be done by drawing all the c edges connecting $(ab)^{2k}$ to $(ab)^{2k+1}a$ inside the loop and drawing all the c edges connecting $(ab)^{2k+1}$ to $(ab)^{2k+2}a$ outside the loop. Thus, the graph is planar; that is, its genus is zero. ■

Lemma 6 *If $T(p_1, p_2, p_3)$ is a right triangle then $G = G(p_1, p_2, p_3)$ is not planar.*

Proof. As in the previous lemma we will demonstrate a subgraph of $G = G(p_1, p_2, p_3)$ isomorphic to a subdivision of $K_{3,3}$. Since T is a right triangle, let $\alpha_2 = \dfrac{\pi}{2}$. Then by Lemma 4 and Corollary 5, the set $\{a, b\}$ generates D_n and the subgraph $Cay(\{a, b\}, D_n)$ is a Hamiltonian cycle of G. Observe that

$p_2 = p_1 + p_3$, and so $n = 2(p_1 + p_3)$ is even.

To determine the structure of the rest of G we must determine how c is related to a and b. Since $\{a, b\}$ generates D_n, any rotation can be expressed as an integer power of ab, and hence ab must be a cyclic generator for the Euclidean rotation subgroup of D_n. This subgroup has order n, and so n is the smallest positive integer such that $(ab)^n = Rot(2\pi)$. Since n is even, it follows that $(ab)^{n/2} = Rot(\pi)$. We have $c = Ref(\pi/2)$ and $a = Ref(0)$. Thus $(ab)^{n/2}a = Rot(\pi)Ref(0) = Ref\left(0 + \frac{1}{2}\pi\right) = c$.

Since n is even, we know that $n \geq 4$. If $n = 4$ then $T = T(2, 1, 1)$ is isosceles so the claim follows from Lemma 5. Otherwise, consider Figure 14, which demonstrates that for $n > 4$ the sets $V_1 = \{e, (ab)^2, (ab)^{n/2+1}\}$ and $V_2 = \{aba, (ab)^{n/2}a, (ab)^{n/2+2}a\}$ can be used as the bipartite sets for a subgraph of G that is a subdivision of $K_{3,3}$. ■

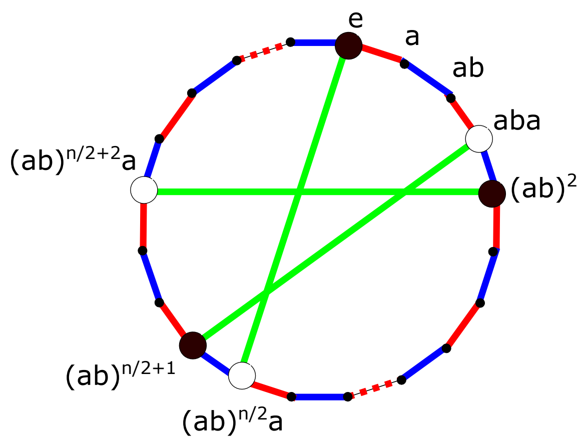

Figure 14: A subgraph of the Cayley graph for a right triangle which is isomorphic to a subdivision of $K_{3,3}$. The three white vertices form a bipartite set; the three black vertices form the other.

Theorem 6 *The genus of $G = G(p_1, p_2, p_3)$ is always 0 or 1. In particular, the genus is zero if and only if $T(p_1, p_2, p_3)$ is isosceles and $n = p_1 + p_2 + p_3$ is even.*

Proof. By Lemma 2, the genus g of $G(p_1, p_2, p_3)$ is at most 1. Next we determine when $g = 0$. Suppose that $g = 0$, and that ρ is a maximal rotation of G. Since all circuits of G have even length of at least 4, it follows that ρ must induce a circuit of length 4. By Lemma 3, this can only occur if $T(p_1, p_2, p_3)$ is right or isosceles. But by Lemma 6, a genus 0 graph cannot arise from a right triangle. Lemma 5 finishes the proof. ■

5 Other Polygons

We observe that Theorem 6 easily extends to any polygon whose sides have no more than 3 distinct slopes.

Corollary 7 *Let P be a rational-angled polygon in the plane and suppose that the edges of P have k distinct slopes (possibly including ∞). If $k \leq 3$ then the Cayley graph arising from P will have genus 0 or 1.*

Proof. Recall that reflections across parallel edges have the same derivative, and that the group D_n generated by reflections across the edges of P actually consists of derivatives of reflections. Therefore, if P has k distinct slopes then each copy of P on its billiards surface will share a side with exactly k other copies of P. Hence, the Cayley graph G arising from P is k-regular. If $k = 3$ then there exists a rational-angled triangle T with the same set of edge slopes, and hence the graph G is the Cayley graph corresponding to T, and Theorem 6 applies. If $k = 2$ then G is a connected 2-regular graph and hence its own Hamiltonian cycle, which has genus 0. ■

So, for example, all trapezoids will yield graphs of genus 0 or 1, as will all "L-shaped" billiard tables.

On the other hand, it follows from Theorem 4 that most rational polygons will have Cayley graphs with genus greater than 1.

Corollary 8 *Let P be a rational-angled polygon in the plane and suppose that the edges of P have k distinct slopes (possibly including ∞). If $k \geq 5$ then the Cayley graph G arising from P will have genus $g \geq 2$.*

Proof. Let G have $2n$ vertices. Then it has $\frac{1}{2}(2nk) = nk$ edges and $2nk$ directed edges. Since each directed edge belongs to exactly one circuit, we have that $A = \frac{2nk}{r(\rho)}$, where $r(\rho)$ is the number of circuits induced by a maximal graph rotation. Hence $r(\rho) = \frac{2nk}{A}$. Substituting into the formula given in Theorem 4, we get:

$$2 - 2g = 2n - nk + \frac{2nk}{A}$$

Solving this for g yields:

$$g = 1 + n\left[\frac{kA - 2k}{2A} - 1\right] \tag{4}$$

Since G is bipartite, each circuit has length at least 4, and so we know that $A \geq 4$. Thus, viewing g as a function of A in Equation 4, g has its minimum value when $A = 4$. Therefore, $g \geq 1 + n\left[\frac{k}{4} - 1\right]$. If $k \geq 5$, then (recalling that g is an integer), we see that $g \geq 2$. ■

6 Future Work

It would be interesting to extend the results of this paper to all rational-angled polygons. Corollary 8 indicates that most rational polygons will correspond to graphs which have genus at least 2, and in fact that there exist corresponding Cayley graphs of arbitrarily high genus; however, we do not yet have an exact formula for genus in the general case.

It would also be interesting to explicitly describe the relationship between the genus of a billiard surface and the genus of its corresponding Cayley graph in terms of the graph rotation defined by the initial drawing of the graph. Intuitively, we hypothesize that the graph tends to have a lower genus because we are allowing it to be drawn differently than it was initially drawn on its billiard surface. That is, a particular drawing of a graph defines a rotation of that graph, but in computing the genus of the graph we consider other possible graph rotations that may induce more circuits (and hence correspond to a drawing on a lower-genus surface).

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